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Delivery system aid and associated systems and methods

Abstract

An endoluminal delivery system including an introducer sheath, a delivery catheter configured to pass through the introducer sheath, an endoluminal device releasably maintained along the delivery catheter at a delivery profile, and a guidewire tube removably received by the endoluminal device, the guidewire tube including an enlarged feature configured to prevent insertion of the guidewire tube into a body of a patient.

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Background/Summary

(1) This application is a continuation of U.S. application Ser. No. 16/965,696, filed Jul. 29, 2020, which is a U.S. 371 Application of International Application PCT/US2018/017037, filed Feb. 6, 2018, which claims the benefit of U.S. Provisional Application No. 62/625,654, filed Feb. 2, 2018, all of which are herein incorporated by reference in their entirety.

FIELD

(1) The present disclosure relates generally to endoluminal delivery systems. More specifically the disclosure relates to endoluminal delivery system aids, and even more specifically, removable guidewire tubes of endoluminal delivery systems.

BACKGROUND

(2) Stents are examples of expandable endoluminal prosthetic devices, which may be used to maintain, open or dilate stenotic lesions in body lumens or to repair an aneurysm or dissection. Vascular disease may occur at a branch or bifurcation in a vessel, such as the aortic bifurcation, common iliac arteries, or other such areas. However, placement and deployment of these prosthetic devices at bifurcations can often be problematic.

(3) One technique is to initially deploy across an aneurysm a main body prosthetic device having one or more sidewall openings. The one or more sidewall openings can be aligned with one or more side branch ostia. One or more additional, prosthetic devices can then be deployed through the main body prosthetic device sidewall opening(s) and into the side branch vessel(s). Procedural complications can be encountered while practicing such techniques, including limitations on accurate placement of the main body prosthetic device (e.g., including alignment of sidewall opening(s) to side branch vessel(s)).

(4) Placement of side branch guidewires through the main body prosthetic device, through the sidewall opening(s) and then into the side branch vessel(s) subsequent to main body prosthetic device deployment can be problematic. For example, difficulties can be encountered if the sidewall

opening(s) are not properly aligned to begin with, and accessing and passing through the sidewall opening(s) can prove challenging. Additionally, delivery systems that include multiple catheters or other features for maintaining side branch guidewire(s) can be overly complex, and may interfere or complicate removal of the main body delivery catheter prior to the delivery of the side branch prosthetic device(s) through the sidewall opening(s) in the main body prosthetic device.

(5) Alternate procedures that place sidewall opening guidewires prior to the main body device deployment can help with aligning guidewire(s) and/or the sidewall opening(s) with the side branch ostium(s), for example. Some methods and systems have been proposed that include pre-loaded side branch opening guidewires (i.e., guidewires that have been loaded into the side branch opening(s) prior to placing the main body prosthetic device at the desired treatment site) utilize guidewire tubes to assist with pre-loading or otherwise guiding guidewires into the side branch opening(s). U.S. Pat. No. 8,267,988, entitled “Side branched endoluminal prostheses and methods of delivery thereof,” issued to Hamer et al., provides some examples of known guidewire tubes and pre-loaded side branch guidewires used for such purposes.

(6) For example, Hamer et al. describe a main body prosthetic device having one or more side branch guidewire lumens, or guidewire tubes, that pass through, or cannulate, the main body prosthetic device and side openings in the main body prosthetic device. One or more guidewires are able to be placed into the main body prosthetic device, and through the side opening(s) by passing the guidewire(s) through the side branch guidewire lumen(s). In some examples, the side branch guidewire lumens are removed prior to advancing the main body device to a desired treatment site in the body. In one method of delivery, as the main body device is advanced, the side opening(s) are self-guided (e.g., by the side branch guidewire(s)) and self-align to the side branch vessel ostium(s). In such examples, the main body device can then be deployed, leaving the side branch guidewire(s) in place. One or more side branch devices can then be advanced along the side branch guidewire(s) through the main body device, through the side wall opening(s) and into the native side branch vessel(s).

SUMMARY

(7) Various examples relate to an endoluminal delivery system including an introducer sheath, a delivery catheter, a first guidewire, an endoluminal device (e.g., a stent, a stent-graft, a vascular filter, a transcatheter valve, or other endoprosthesis), a second guidewire, and a guidewire tube. The guidewire tube includes an enlarged feature and is configured to be removably received by a portion of the endoluminal device.

(8) According to one example (“Example 1”), an endoluminal delivery system includes an introducer sheath having a proximal end, a distal end, a lumen extending between the proximal end and the distal end, and a proximal opening at the proximal end providing access to the lumen. The endoluminal delivery system also includes a delivery catheter configured to be passed through the proximal opening into the lumen of the introducer sheath. The endoluminal delivery system also includes an endoluminal device releasably maintained along the delivery catheter at a delivery profile. The delivery profile is configured so that the endoluminal device can be passed through the proximal opening into the lumen of the introducer sheath while the endoluminal device is maintained in the delivery profile. The endoluminal device has an inner lumen, a first opening providing access to the inner lumen, and a second opening providing access to the inner lumen. The endoluminal delivery system also includes a guidewire tube removably received by the endoluminal device so that the guidewire tube is removable from the endoluminal device when the endoluminal device is in the delivery profile. The guidewire tube passes through the first opening, into the lumen, and out of the second opening of the endoluminal device. The guidewire tube has a first end, a second end, and a lumen configured to receive a guidewire. The guidewire tube also includes an enlarged feature configured to selectively prevent insertion of the guidewire tube through the proximal opening and into the lumen of the introducer sheath.

(9) According to another example (“Example 2”) further to Example 1, the proximal opening into

the lumen of the introducer sheath is defined by a hemostatic valve.

(10) According to another example (“Example 3”) further to any of Examples 1 and 2, the endoluminal delivery system also includes a guidewire received in the lumen of the guidewire tube.

(11) According to another example (“Example 4”) further to any of Examples 1 to 3, the endoluminal delivery system also includes a constraint releasably maintaining the endoluminal device in the delivery configuration.

(12) According to another example (“Example 5”) further to any of Examples 1 to 4, the enlarged feature includes a portion of the guidewire tube that defines an enlarged outer profile relative to a remaining portion of the guidewire tube.

(13) According to another example (“Example 6”) further to any of Examples 1 to 5, the enlarged feature includes at least one of a bead, a ring, a barb, a knob, and a catch.

(14) According to another example (“Example 7”) further to any of Examples 1 to 6, the enlarged feature is substantially teardrop shaped.

(15) According to another example (“Example 8”) further to any of Examples 1 to 7, the enlarged feature is positioned at a proximal end of the guidewire tube.

(16) According to another example (“Example 9”) further to any of Examples 1 to 8, the enlarged feature is positioned at a distal portion of the guidewire tube.

(17) According to another example (“Example 10”) further to any of Examples 1 to 9, the enlarged feature defines a maximum diametric dimension that is greater than a maximum diametric dimension of adjacent portions of the guidewire tube by at least 10%, 20%, 30%, 40%, 50%, 60%, 100%, 200%, 300%, or more.

(18) According to another example (“Example 11”), an endoluminal delivery system includes an introducer sheath having a proximal end, a distal end, and a lumen extending between the proximal end and the distal end. The endoluminal delivery system also includes a delivery catheter. The endoluminal delivery system also includes a constraint having proximal end, a distal end, a wall extending from the proximal end to the distal end to form a lumen, and at least one opening in the wall. The endoluminal delivery system also includes a guidewire tube having a first end, a second end, and a lumen. The guidewire tube extends from the at least one opening through the lumen of the constraint. The guidewire tube has an increased profile section configured to prevent insertion of the guidewire tube into the introducer sheath.

(19) According to another example (“Example 12”), a method for using an endoluminal delivery system of any of Examples 1 to 10 includes delivering a guidewire to a desired treatment site within the body of a patient. The method also includes inserting the guidewire through the guidewire tube so that the guidewire extends through a lumen of the guidewire tube. The method also includes removing the guidewire tube from the endoluminal delivery system. The method also includes inserting the endoluminal delivery system into the body of a patient and delivery the endoluminal device to the desired treatment site within the body of the patient.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments, and together with the description serve to explain the principles of the disclosure.

(2) FIGS. 1A and 1B show an endoluminal delivery system, according to some embodiments.

(3) FIG. 2 shows a delivery catheter with an endoluminal device maintained along the delivery catheter, and a guidewire tube, according to some embodiments.

(4) FIG. 3 shows a guidewire tube including an enlarged feature, according to some embodiments.

(5) FIGS. 4a-d shows additional features of a guidewire tube including an enlarged feature,

according to some embodiments.

(6) FIG. 5 is a schematic view of a delivery catheter having a guidewire tube therethrough, according to some embodiments.

(7) FIG. 6 is a schematic view of a delivery catheter with an endoluminal device maintained along the delivery catheter and a constraint, according to some embodiments.

(8) FIGS. 7-11 show a method for using an endoluminal delivery system, according to some embodiments.

DETAILED DESCRIPTION

(9) FIG. 1A shows a schematic view of an endoluminal delivery system **10**, according to some embodiments. As shown, the endoluminal delivery system **10** includes an introducer sheath **20**, a delivery catheter **40**, a first guidewire **50**, an endoluminal device **60**, a second guidewire **70**, and a guidewire tube **80**. In various examples, the endoluminal device **60** is maintained on the delivery catheter **40** in a delivery configuration for endoluminal delivery, including passage of the endoluminal device **60** through the introducer sheath **20**. For reference, a delivery configuration is meant to refer to a diametrically compacted, or simply compacted profile whereas a deployed configuration is meant to refer to a diametrically enlarged, or simply enlarged profile. As described in further detail, the guidewire tube **80** includes an enlarged feature **100** and is configured to be removably received by a portion of the endoluminal device **60** (e.g., prior to delivering the endoluminal device **60** to a desired location in the body of a patient). For reference, the enlarged feature **100** is optionally configured as a stop, a removal aid, an identifier, or combinations thereof.

(10) In various examples, the endoluminal delivery system **10** facilitates deployment of the endoluminal device **60**, which may include a stent, a graft, a stent-graft, a vascular filter, a transcatheter valve, or other endoprosthesis, at a desired treatment location within the body of a patient. In some examples, the desired treatment location may be an area of vascular bifurcation such as the aortic arch branches (arteries of the head, arms, and hands), lower branches of the aorta (celiac), renals, mesenterics, iliacs, the femoral, and/or the lower extremities (legs and feet).

(11) As shown in FIG. 1A, the introducer sheath **20** has a proximal end **22**, a distal end **24**, and a lumen **26** extending between the proximal end **22** and the distal end **24**. The introducer sheath **20** may have an inner diameter and a length that depend on the desired treatment location. For example, the inner diameter (e.g., at proximal opening **28**) may be between about 3 mm and about 10 mm, and the length may be between about 10 mm and about 30 cm, between about 30 cm and about 70 cm, or alternatively may be over 70 cm.

(12) In some embodiments, the introducer sheath **20** may be substantially straight, tapered, stepped, or any combination thereof. For example, the introducer sheath **20** can have an inner diameter (e.g., defined at proximal opening **28**) that is substantially constant along the length of the lumen **26** (e.g., from the proximal end **22** to the distal end **24**). In another example, the inner diameter can vary along the length of the lumen **26**. For example, the inner diameter may be smaller at the proximal end **22** and increase to a larger diameter at the distal end **24**, or vice versa.

(13) In some embodiments, proximal opening **28** at the proximal end **22** provides access to the lumen **26**. There may also be a distal opening **30** at the distal end **24** such that, in some embodiments, the delivery catheter **40** can be inserted into the introducer sheath **20** through the proximal opening **28** and exit through the distal opening **30** of the introducer sheath **20**.

(14) As indicated generally in FIG. 1A, the introducer sheath **20** has a proximal portion **28a** defined by one or more features of the introducer sheath **20**, which, in turn, defines the proximal opening **28** of the introducer sheath **20**. In some examples, the proximal portion **28a** may be tubular projection (e.g., of sheath material), a hemostatic valve that defines the proximal opening **28**, or other feature of the introducer sheath **20**, such as a luer fitting. Though the proximal portion **28a** is indicated generally in FIG. 1A by broken lines, and may include a variety of features associated with introducer sheath designs, one specific example of an introducer sheath including a hemostatic valve for the proximal portion **28a** is that sold under the trade name, “DrySeal Flex Introducer

Sheath,” available from W.L. Gore & Associates Inc. For example, FIG. 1B provides an example of the introducer sheath **20** with a valve portion **200** defining the proximal portion **28a** of the introducer sheath **20**, including the proximal opening **28**, as well as a sheath portion **202** extending from the valve portion **200**. From the more schematic representation of FIG. 1A and the associated description, it should be apparent that an introducer sheath having a proximal portion **28a** formed by another feature (e.g., a luer fitting) and defining the proximal opening **28** is also contemplated.

(15) As seen in FIG. 2, the delivery catheter **40** has a proximal end **42**, a distal end **44**, and an inner lumen **46**. As discussed above, the delivery catheter **40** is configured to be passed through the proximal opening **28** and into the lumen **26** of the introducer sheath **20**. Thus, the delivery catheter **40** should be sized such that it can be passed through the lumen **26** of the introducer sheath **20**. For example, the delivery catheter **40** can have a diameter ranging from about 1 mm to 20 mm, 2 mm to 15 mm, or 2 mm to 6 mm at a portion of the delivery catheter **40** configured to pass through the introducer sheath **20**. Although some examples are provided, any variety of diameters is contemplated.

(16) In some embodiments, the delivery catheter **40** has a proximal portion **47** extending between the proximal end **42** and a point located longitudinally between the proximal end **42** and the distal end **44**. The delivery catheter **40** may also have a distal portion **48** extending between the distal end **44** and a point located longitudinally between the proximal end **42** and the distal end **44**. Both the proximal portion **47** and the distal portion **48** can have diameters similar to those listed above for the delivery catheter **40**.

(17) The delivery catheter **40** generally has a length commensurate with delivering an endoluminal device to a desired treatment site. Thus, the delivery catheter **40** can have a length suitable to deliver any of a variety of prosthetic devices. The length of the delivery catheter **40**, including both the proximal portion **47** and distal portion **48**, can vary from about 20 cm to over 100 cm depending on the desired treatment location.

(18) Similar to the introducer sheath **20** discussed above, in some embodiments, the delivery catheter **40** may be substantially straight, tapered, stepped, or any combination thereof. For example, the delivery catheter **40** can have a diameter that is substantially constant along the length of the inner lumen **46** (e.g., from the proximal end **42** to the distal end **44**). In another example, the inner diameter can vary along the length of the inner lumen **46**. For example, the diameter may be smaller at the proximal end **42** and may increase to a larger diameter at the distal end **44**, or vice versa.

(19) The delivery catheter **40** can comprise conventional medical grade materials such as nylon, polyacrylamide, polycarbonate, polyethylene, polyformaldehyde, polymethylacrylate, polypropylene, polytetrafluoroethylene, polytrifluorochlorethylene, polyvinylchloride, polyurethane, elastomeric organosilicon polymers, Pebax® polyether block amide, and metals such as stainless steels and nitinol.

(20) In some embodiments, the endoluminal device **60**, also referred to herein as an implantable device or simply a device, is mounted on the delivery catheter **40**, and specifically by the distal portion **48**. In one embodiment, the device **60** is maintained at a delivery diameter such that the device **60** can be passed through the lumen **26** of the introducer sheath **20** and moved to the desired treatment location.

(21) The device **60** has a proximal end **62**, a distal end **64**, and an inner lumen **66**. The device **60** may also include a plurality of openings **67** (three are shown in FIG. 2, though any number is contemplated) providing access to the inner lumen **66**. In some examples, the device **60** includes a first opening **68a** (e.g., an end opening or a side opening) providing access to the inner lumen **66** and a second opening **68b** (e.g., an end opening or a side opening) providing access to the inner lumen **66**, and a third opening **68c** (e.g., an end opening or a side opening) providing access to the inner lumen **66**. Any of the first opening **68a**, the second opening **68b**, and/or the third opening **68c** may be sized to fit the guidewire tube **80** therethrough.

(22) As shown, the first opening **68a** is an end opening located at the proximal end **62** (FIG. 2).

However, the first opening **68a** may be located at any point between the proximal end **62** and the distal end **64** of the device **60**, including proximate the proximal end **62** of the device **60**, proximate the distal end **64** of the device **60**, or intermediate the proximal end **62** and the distal end **64** of the device **60**. In one example, the second opening is located at a point proximate the distal end **64** of the device **60** (FIG. 2). However, the second opening **68b** is optionally located at any point between the proximal end **62** and the distal end **64** of the device **60**, including proximate the proximal end **62** of the device **60**, intermediate the proximal end **62** and the distal end **64** of the device **60**, or at an end of the device **60**. In some embodiments, the device **60** may have more or fewer than three openings. For example, the device **60** can have a first opening **68a**, a second opening **68b**, and any number of subsequent openings. The number of openings in any particular embodiment may depend on a variety of factors, including the desired treatment location, the number of access openings desired into the inner lumen (e.g., to receive a desired number of branch devices), or other additional or alternative considerations.

(23) In some embodiments, the device **60** is a stent, a graft, a combination thereof, or another type of prosthetic device altogether. As is generally understood, stents can be used alone or in combination with graft materials. Stents can be configured on the external or internal surface of a graft or may be incorporated into the internal wall structure of a graft. Grafts can have various configurations and can be fabricated, for example, from tubes, sheets or films formed into tubular shapes, woven or knitted fibers or ribbons or combinations thereof. Graft materials can include conventional medical grade materials such as nylon, polyester, polyethylene, polypropylene, polytetrafluoroethylene, polyurethane and elastomeric organosilicon polymers.

(24) In some embodiments, the stent, graft, or other prosthetic device can be either self-expanding or balloon expandable. Typically, a self-expanding device will include at least one shape memory material, such as nitinol. Suitable stent materials include, in addition to nitinol, for example, metallic, polymeric or natural materials and can comprise conventional medical grade materials such as nylon, polyacrylamide, polycarbonate, polyethylene, polyformaldehyde, polymethylmethacrylate, polypropylene, polytetrafluoroethylene, polytrifluorochlorethylene, polyvinylchloride, polyurethane, elastomeric organosilicon polymers; metals such as stainless steels, cobalt-chromium alloys and nitinol and biologically derived materials such as bovine arteries/veins, pericardium and collagen. Stents can also comprise bioresorbable materials such as poly(amino acids), poly(anhydrides), poly(caprolactones), poly(lactic/glycolic acid) polymers, poly(hydroxybutyrates) and poly(orthoesters).

(25) In some embodiments, the guidewire tube **80** (FIG. 3) has a first end **82**, a second end **84**, and a lumen **86**, a proximal portion **87**, and a distal portion **88**. The first end **82** and the second end **84** may be open to provide access to the lumen **86** and receive a guidewire therethrough. In some embodiments, only one of the first end **82** or the second end **84** may be open such that the other, opposite end is closed to permit access to the lumen **86** with a guidewire, but not to allow the guidewire to pass all the way through the lumen **86**.

(26) As referenced above, the guidewire tube **80** is generally sized, shaped, and otherwise configured to fit a guidewire therethrough. In some embodiments, the guidewire tube **80** can have an inner diameter ranging from about 0.1 mm to about 2 mm, 0.2 mm to about 1.5 mm, or 0.3 mm to about 1 mm, or any range including any of the foregoing ranges. The guidewire tube may have a wall thickness ranging from about 0.05 mm to about 1 mm, 0.06 mm to about 0.5 mm, or 0.08 mm to about 0.3 mm, or any range including any of the foregoing ranges. In some embodiments, the wall thickness may be substantially uniform along a length of the guidewire tube **80**. For example, the wall thickness may be substantially the same from the first end **82** to the second end **84**. In other embodiments, the wall thickness may be non-uniform along the length of the guidewire tube **80** such that the wall thickness varies between the first end **82** and the second end **84**.

(27) The length of the guidewire tube **80** can vary depending on the desired treatment location. For example, the length of the guidewire tube **80** may be tailored for a particular prosthetic device. In

some embodiments, the guidewire tube **80** may be significantly shorter than the overall catheter length and may be slightly longer than the device **60**. For example, the guidewire tube **80** may have a length ranging from about 1 cm to about 30 cm, 2 cm to about 20 cm, or 4 cm to about 15 cm, or any range including any of the foregoing ranges.

(28) The guidewire tube **80** can comprise suitable medical grade materials similar to those described above for the delivery catheter **40**. Examples of such materials include conventional medical grade materials such as nylon, polyacrylamide, polycarbonate, polyethylene, polyformaldehyde, polymethylmethacrylate, polypropylene, polytetrafluoroethylene, polytrifluorochlorethylene, polyvinylchloride, polyurethane, polyether ether ketone (PEEK), elastomeric organosilicon polymers, and metals such as stainless steel and nitinol. In some embodiments, the material may be sufficiently translucent so that a guidewire can be visualized by the practitioner as the guidewire is advanced through the guidewire tube **80**.

(29) In some embodiments, one or more additional guidewire tubes similar to the guidewire tube **80** may be used. Additional guidewire tubes (not shown) may be useful, for example, when treating a bifurcated area within the body of a patient (or an area with multiple branch ostia) using an endoprosthesis with multiple sidewall openings.

(30) In some embodiments, the guidewire tube **80** includes an enlarged feature **100** (FIG. 3). The enlarged feature **100**, also referred to as a removal aid, an identifier, or any combination thereof, may serve as a visual reminder to the practitioner to remove the guidewire tube **80** prior to insertion of the device **60** and/or delivery catheter **40** into the introducer sheath **20**. The enlarged feature **100** may also aid in removal of the guidewire tube **80** by providing an enlarged profile for the practitioner to grip when pulling, sliding, or otherwise removing the guidewire tube **80**. In yet another embodiment, the enlarged feature **100** may prevent removal of the guidewire tube **80** from the device **60** and/or delivery catheter **40** in one direction, while enabling removal in the opposite direction. In some examples, the enlarged feature **100** is a diametrically enlarged portion of the guidewire tube **80**. For example, the enlarged feature optionally defines a maximum diametric dimension that is greater than that of adjacent portions of the guidewire tube **80** by at least 10%, 20%, 30%, 40%, 50%, 60%, 100%, 200%, 300%, or any range including any of the foregoing ranges. Although some examples have been provided, any enlarged profile portion of any desirable dimension is contemplated.

(31) In some embodiments, the enlarged feature **100** is located at the first end **82** of the guidewire tube **80** (FIG. 3). However, the enlarged feature **100** may be optionally located at the second end **84** of the guidewire tube **80**, along the proximal portion **87**, along the distal portion **88**, or along any other point along the guidewire tube **80** between the first end **82** and the second end **84**.

(32) FIGS. 4a-d show example alternative configurations of the guidewire tube **80** and the enlarged feature **100**. As shown, the enlarged feature **100** may be any one of a bead, a ring, a crimp, a bulge, a barb, a knob, a foot, a catch, or any other feature that provides an enlarged outer profile to prevent insertion of the device **60** and/or delivery catheter **40** into the introducer sheath **20** prior to removal of the guidewire tube **80**. For example, in one embodiment, the enlarged feature **100** has a diameter that is larger than the diameter of the guidewire tube **80**. In other embodiments, the enlarged feature **100** may have a different shape, color, or profile than the guidewire tube **80**. For example, the enlarged feature **100** may be substantially round, teardrop-shaped, L-shaped, hook-shaped, or any other shape that prevents insertion of the guidewire tube **80** into the introducer sheath **20**.

(33) In some examples, multiple guidewire tubes are provided (e.g., as part of a single delivery system, surgical kit, or other arrangement), where each of the multiple guidewire tubes includes an enlarged feature, such as the enlarged feature **100**. In such instances, differing enlarged features configurations may be used (e.g., different size, shape, color, or other differentiator) to differentiate between multiple guidewires, guidewire tubes, or various other features.

(34) In some embodiments, the enlarged feature **100** comprises the same material as the guidewire

tube **80**. However, the enlarged feature **100** may also comprise a different material than the guidewire tube **80**. For example, the enlarged feature **100** may comprise ultra-violet (UV) curable adhesives and sealants, thermoset plastics, thermoplastics, compliant polymers, extrusions such as bump extrusions, metal ferrules, and various overmolded components.

(35) The surface of the enlarged feature **100** may vary depending on various factors, including desired use and/or practitioner preference. In some embodiments, the enlarged feature **100** is substantially smooth. However, in other examples, the surface may comprise indentations, lines, dimples, crosshatches, or any other such texture or combination of textures.

(36) Referring to FIG. 2, the device **60** is coupled to the delivery catheter **40** at the distal portion **48** of the delivery catheter **40**. In some examples, the guidewire tube **80** is removably received through the first opening **68a** of the device **60**. However, the guidewire tube **80** can also be received by the second opening **68b**, the third opening **68c**, or any other additional opening in the device **60**. In some embodiments, the guidewire tube **80** may enter through one opening and exit through another opening.

(37) Any additional guidewire tubes necessary for completion of the desired procedure may extend, for example, from the first opening **68a** to the second opening **68b**, the third opening **68c**, or other, additional openings within the device **60**.

(38) In another optional configuration (FIG. 5), the guidewire tube **80** may be removably received by one or more portions of the delivery catheter **40**. For example, the guidewire tube **80** may enter through the proximal end **42** or the distal end **44** of the delivery catheter **40**. In some embodiments, the guidewire tube **80** may enter through the proximal end **42** or the distal end **44** and exit through a first opening **49** in the delivery catheter **40**. However, it is contemplated that the guidewire tube **80** may enter and exit through any combination of openings in the delivery catheter **40**. Similarly as discussed above, any number of additional guidewire tubes necessary for completion of the desired procedure are contemplated.

(39) In FIG. 5, the enlarged feature **100** is shown at the first end **82** of the guidewire tube **80**. However, as discussed above, the enlarged feature **100** can be located at the first end **82**, the second end **84**, or at any other point between the first end **82** and the second end **84**.

(40) In some embodiments, the enlarged feature **100** is adhesively attached to the guidewire tube **80**. The enlarged feature **100** must be attached sufficiently to withstand a removal force (i.e. pulling, sliding, etc.). For example, the enlarged feature **100** should be adequately adhered/attached/connected to the guidewire tube **80** so as not to deform, move from its original position, or detach from the guidewire tube **80** during removal of the guidewire tube **80** from the device **60**. Therefore, in various embodiments, the enlarged feature **100** may have an attachment force of at least about 1 lbf, at least about 4 lbf, at least about 7 lbf, or at least about 10 lbf. In other embodiments, the attachment force is between about 1 to 12 lbf, between about 6 to 12 lbf, and/or between about 8 to 12 lbf, or any range including any of the foregoing ranges.

(41) In FIG. 6, the device **60** is shown in a compressed state (e.g., at a delivery profile) at the distal portion **48** of the delivery catheter **40**. In one embodiment, the device **60** is maintained in the compressed state by a constraint **120**.

(42) The constraint **120** includes a proximal end **124**, a distal end **126**, and a lumen **129**. In some embodiments, the constraint **120** is a sleeve, a sheath, or a cover capable of compressing the device **60** to the appropriate delivery profile, also referred to herein as the delivery diameter. For example, the device **60** may be compressed to a delivery diameter ranging from about 3 mm to 13 mm or up to about 110 mm, or any range including any of the foregoing ranges, or other value as desired.

(43) The constraint **120** may be disposed over all or a portion of the device **60**. In some embodiments, the constraint **120** is laced together by a deployment cord **122**, which may form a generally longitudinal seam along the constraint **120**.

(44) In some embodiments, the constraint **120** includes a first opening **129** oriented perpendicular to the longitudinal seam. The first opening **129** may provide an exit point for the guidewire tube **80**

to exit the delivery catheter **40** and constraint **120**. However, in other embodiments, the constraint **120** may not include a first opening **129**. For example, the guidewire tube **80** may exit through a stitch line of the longitudinal seam. Details relating to constraining sheath materials, sheath methods of manufacture and main body compression techniques can be found in, for example, U.S. Pat. No. 6,352,561 to Leopold et al., and U.S. Pat. No. 6,551,350 to Thornton et al.

(45) FIGS. **7-11** show a method of using the endoluminal delivery system **10**. As shown in FIG. **7**, the first guidewire **50** is passed through the introducer sheath **20** (not shown) and placed into a main body lumen **150** and the second guidewire **70** is placed into a branch lumen **152**. The first guidewire **50** is then threaded proximally through the distal end **44** and into the inner lumen **46** of the delivery catheter **40**. The second guidewire **70** is threaded into the removable guidewire tube **80**. Typical guidewire diameters include, for example, 0.089 cm (0.035 inches) and 0.36 cm (0.014 inches). However, it should be understood that other size guidewires can be used depending on the particular procedure, desired treatment location, and/or the practitioner's preferences.

(46) FIG. **8** shows the first guidewire **50** and the second guidewire **70** threaded through the delivery catheter **40** and the guidewire tube **80**, respectively. In FIG. **8**, both the first end **82** and the second end **84** of the guidewire tube **80** are open such that the second guidewire **70** extends out of both ends. However, in other embodiments, the first end **82** of the guidewire tube **80** may be closed such that it provides a stopping point for the second guidewire **70** such that the second guidewire **70** does not extend out of the first end **82**.

(47) Once the second guidewire **70** is placed within the guidewire tube **80**, the guidewire tube **80** can be removed from the device **60** proximally in a direction shown by the arrows in FIG. **9**. In another configuration, the guidewire tube **80** may be removed distally in the opposite direction of the arrows. As discussed above, the enlarged feature **100** may aid in removal of the guidewire tube **80** by providing an enlarged portion or handhold for the practitioner to grip. The enlarged feature **100** may also provide a visual reminder to the practitioner to remove the guidewire tube **80** prior to insertion of the delivery catheter **40** and device **60** into the body of the patient. Further, the enlarged feature **100** may facilitate removal in only one direction (i.e., either proximally in the direction denoted by the arrows in FIG. **9** or distally in the opposite direction of the arrows in FIG. **9**).

(48) Once the guidewire tube **80** is removed, the delivery catheter **40** with device **60** maintained at the distal portion **48** is advanced along the first guidewire **50** to the desired treatment location until the second opening **68b** is aligned with the branch lumen **152**. FIG. **10** shows deployment of the device **60** within the main lumen **150**. As shown, the second opening **68b** is aligned with the branch lumen **152** and the second guidewire **70** remains extended through the second opening **68b** and the branch lumen **152**.

(49) After deployment of the device **60** within the main lumen **150**, a branch device **160** (FIG. **11**) may be advanced along the second guidewire **70** to the branch lumen **152**. In some embodiments, the branch device **160** can be a stent, graft, or other prosthetic device similar to the device **60**. The branch device **160** can be compressed in a delivery profile and can include a constraint similar to the device **60**, as discussed above. Further, the branch device **160** can include a branch deployment cord **162** (FIG. **10**), which can be released once the branch device **160** is in the desired location within the branch lumen **152**. In some embodiments, the branch device **160** may be self-expandable or balloon expandable. In some examples (e.g., self-expandable branch device examples), when the branch deployment cord **162** is pulled, the branch device **160** expands to fit the branch lumen **152**, as shown in FIG. **11**.

(50) As discussed above, the endoluminal delivery system **10** may include any number of openings, guidewire tubes, guidewires, and/or branch devices for the desired application. This may depend, for example, on the desired treatment location, the number of branch devices required, and/or the size or number of aneurysms to be treated. Deployment of any subsequent branch devices may be realized in a similar manner as described above.

(51) Persons skilled in the art will readily appreciate that various aspects of the present disclosure

can be realized by any number of methods and apparatus configured to perform the intended functions. It should also be noted that the accompanying drawing figures referred to herein are not necessarily drawn to scale, but may be exaggerated to illustrate various aspects of the present disclosure, and in that regard, the drawing figures should not be construed as limiting.

(52) The invention of this application has been described above both generically and with regard to specific embodiments. It will be apparent to those skilled in the art that various modifications and variations can be made in the embodiments without departing from the scope of the disclosure. Thus, it is intended that the embodiments cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. An endoluminal delivery system, comprising: an introducer sheath having a proximal end, a distal end, a lumen extending between the proximal end and the distal end, and a proximal opening at the proximal end providing access to the lumen; a delivery catheter configured to be passed through the proximal opening into the lumen of the introducer sheath; an endoluminal device releasably maintained along the delivery catheter at a delivery profile that is configured for the endoluminal device to be passed through the proximal opening into the lumen of the introducer sheath while the endoluminal device is maintained in the delivery profile, the endoluminal device having an inner lumen, a first opening providing access to the inner lumen, and a second opening providing access to the inner lumen; a plurality of guidewire tubes removably received by the endoluminal device such that the guidewire tubes are removable from the endoluminal device when the endoluminal device is in the delivery profile, wherein each of the guidewire tubes passes through the first opening of the endoluminal device into the inner lumen of the endoluminal device and out of the second opening of the endoluminal device, wherein each of the guidewire tubes has a first end, a second end, a lumen configured to receive a guidewire, and an enlarged feature configured to selectively prevent insertion of the guidewire tube through the proximal opening of the introducer sheath into the lumen of the introducer sheath, wherein the enlarged feature of each guidewire tube has a different color than the other enlarged features of the other guidewire tubes.
2. An endoluminal delivery system, comprising: an introducer sheath having a proximal end, a distal end, a lumen extending between the proximal end and the distal end, and a proximal opening at the proximal end providing access to the lumen; a delivery catheter configured to be passed through the proximal opening into the lumen of the introducer sheath; an endoluminal device releasably maintained along the delivery catheter at a delivery profile that is configured for the endoluminal device to be passed through the proximal opening into the lumen of the introducer sheath while the endoluminal device is maintained in the delivery profile, the endoluminal device having an inner lumen, a first opening providing access to the inner lumen, and a second opening providing access to the inner lumen; a plurality of guidewire tubes removably received by the endoluminal device such that the guidewire tubes are removable from the endoluminal device when the endoluminal device is in the delivery profile, wherein each of the guidewire tubes passes through the first opening of the endoluminal device into the inner lumen of the endoluminal device and out of the second opening of the endoluminal device, wherein each of the guidewire tubes has a first end, a second end, a lumen configured to receive a guidewire, and an enlarged feature configured to selectively prevent insertion of the guidewire tube through the proximal opening of the introducer sheath into the lumen of the introducer sheath, wherein the enlarged feature of each guidewire tube has a surface texture that is different from the other enlarged features of the other guidewire tubes.
3. The system of claim 2, wherein the surface texture includes one or more of: a smooth texture, an indented texture, a lined texture, a dimpled texture, or a crosshatched texture.
4. An endoluminal delivery system, comprising: an introducer sheath having a proximal end, a

distal end, a lumen extending between the proximal end and the distal end, and a proximal opening at the proximal end providing access to the lumen; a delivery catheter configured to be passed through the proximal opening into the lumen of the introducer sheath; an endoluminal device releasably maintained along the delivery catheter at a delivery profile that is configured for the endoluminal device to be passed through the proximal opening into the lumen of the introducer sheath while the endoluminal device is maintained in the delivery profile, the endoluminal device having an inner lumen, a first opening providing access to the inner lumen, and a second opening providing access to the inner lumen; a plurality of guidewire tubes removably received by the endoluminal device such that the guidewire tubes are removable from the endoluminal device when the endoluminal device is in the delivery profile, wherein each of the guidewire tubes passes through the first opening of the endoluminal device into the inner lumen of the endoluminal device and out of the second opening of the endoluminal device, wherein each of the guidewire tubes has a first end, a second end, a lumen configured to receive a guidewire, and an enlarged feature configured to selectively prevent insertion of the guidewire tube through the proximal opening of the introducer sheath into the lumen of the introducer sheath, wherein the enlarged feature of each guidewire tube is made of a material that is different from the other enlarged features of the other guidewire tubes.

5. The system of claim 4, wherein the material includes one or more of: an ultra-violet (UV) curable adhesive, an UV curable sealant, a thermoset plastic, a thermoplastic, a compliant polymer, an extrusion, a metal ferrule, or an over-molded component.
